A Change in Observation Is Not a Change in Normality:
Acquisition-Aware Anomaly Detection

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Abstract

Visual anomaly detection treats a large feature residual as a defect. That reading fails when focus, exposure, vibration, noise, or compression changes the image and the object does not. ARAD-Compact scores every official MVTec-AD and VisA test image once clean and once under one stratified acquisition recipe, keeping the label and the defect mask. A training-free detector stores mean multi-layer DINOv2 patch descriptors from a few clean normal images and compares each query patch by squared nearest-neighbor distance. Image detection averages the upper tail of that field. Localization estimates a locally supported field on the feature grid before upsampling. The two readouts share the match and do not share the spatial operator. On clean data the detector matches strong few-shot baselines. Under ARAD-Compact it retains more image- and pixel-level accuracy, because spatial pooling should not enter the image score, a PCA subspace of the same descriptors drops once the observation changes, and encoder scale matters more under shift than on clean images.

1 Motivation and contributions

Few-shot visual anomaly detection asks whether an image, and which of its regions, depart from a handful of normal references. Patch memories, windowed vision–language scores, prompt-based classifiers, and subspace residuals now rank defects accurately on clean MVTec-AD and VisA [1–5]. They share one reading of the residual: a test patch is anomalous when it lies far from the normal set. That reading holds when the object is the only thing that changes. On a line, focus, exposure, vibration, sensor noise, and compression change the image of a physically unchanged part, and the same change hits normal and defective queries. A large residual may then be a defect, an acquisition effect, or both. Figure 1 states the two decisions a robustness claim must get right together. On a shifted normal sample the detector should not invent a defect. On a shifted defective sample it should still mark the defect. A score that absorbs every imaging change can also absorb the defect, and a score that flags every feature change will alarm on blur and exposure.

Clean benchmarks hide the ambiguity, because official queries are acquired under the same conditions as the references. Domain-shift suites change the data distribution. They do not keep the physical sample fixed, apply one explicit recipe, and score the clean and shifted images as a pair.

ARAD-Compact is built from the official test splits of MVTec-AD [6] and VisA [7]. Each source receives exactly one stratified recipe. The clean image, the class label, and the defect mask stay attached to that source. Recipes cover blur (isotropic, anisotropic, defocus, motion), photometric and JPEG effects (under- and over-exposure, sensor noise, JPEG), and mixed compounds, at five severity tiers with shares 10:20:40:20:10. Each single-effect recipe takes 6.25% of the panel and each mixed recipe 8.3% (25% blur, 25% photometric/JPEG, 50% mixed), following a practical degradation model [8] with one operator order. The frozen panel has 1725 MVTec-AD queries and 2162 VisA queries. Any image detector can be reported as Clean | Shift on the same sources. Shift is this mixed panel, not an equal average over a corruption×severity grid. The detector does not name the recipe and does not add shifted images to normality.

The contributions are as follows.

- Acquisition robustness is a paired decision: the same source is scored clean and under one controlled acquisition change, for normal and defective samples.
- ARAD-Compact assigns one stratified recipe to each official MVTec-AD and VisA test image and keeps the clean image, the label, and the mask.
- A training-free mean nearest-neighbor detector shares patch matching across tasks and uses separate spatial readouts, so map regularization does not rewrite the image score.
- Under matched references, the split keeps the localization gain of spatial pooling, nearest neighbors are more stable under ARAD-Compact than a high-variance PCA subspace of the same features, and encoder scale separates methods more under shift than on clean data. Extra shots raise the clean score more than they close the shift gap.

2 Method

Detection depends on the strongest departures from normal appearance. Localization also needs spatially consistent scores. A frozen encoder and a clean reference bank produce one patch-distance field. Detection aggregates its upper

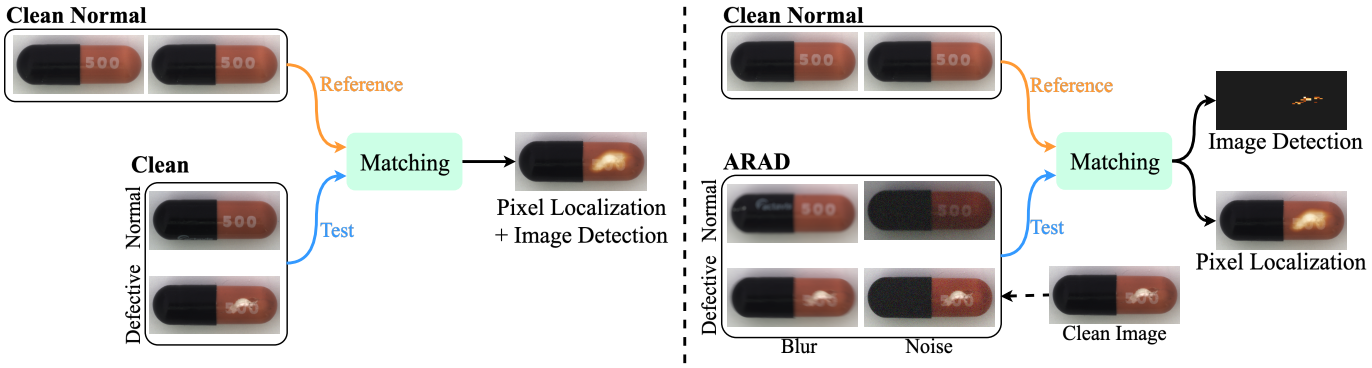

Figure 1: A change in observation is not a change in normality. The same physical sample is judged once clean and once after a known acquisition change. The detector should stay quiet on a shifted normal sample and should still mark a defect when the defect and the acquisition change co-occur.

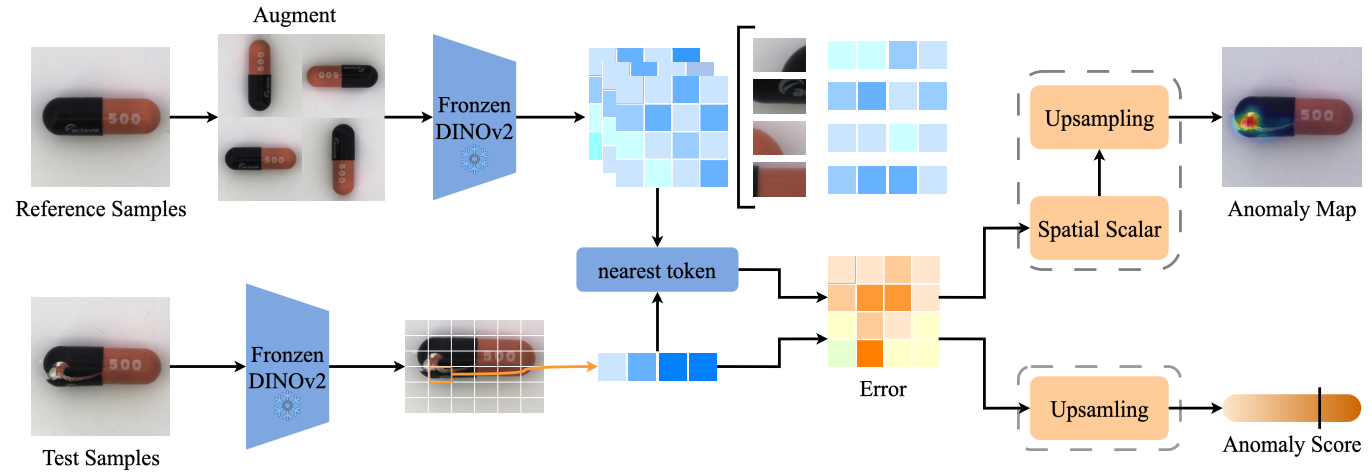

Figure 2: Clean normal references and their geometric views form a memory of mean intermediate-layer patch descriptors. Each query patch retrieves its nearest normal token. Detection upsamples the squared-distance field and averages its top 1%. Localization estimates a spatially supported field on the feature grid before upsampling.

tail after upsampling. Localization estimates a locally supported field before upsampling. Figure 2 shows the shared match and the two readouts.

Let $\mathcal{X}_{\text{ref}} = \{x_i\}_{i=1}^{K}$ be K clean normal images. A frozen DINOv2 encoder [9] yields a patch feature $f_\ell(x, p)$ at block ℓ and location p. The stored descriptor averages a fixed set of blocks $\mathcal{L}$,

$$z(x, p) = \frac{1}{|\mathcal{L}|} \sum_{\ell \in \mathcal{L}} f_\ell(x, p). \tag{1}$$

For each reference, a prescribed set of geometric views $\mathcal{T}_i$ (including the identity) is encoded the same way. The category memory $\mathcal{M} = \{d_j\}$ is the union of those descriptors. K counts distinct source images, and no shifted or anomalous view enters $\mathcal{M}$.

A query patch retrieves its nearest memory entry by squared Euclidean distance,

$$a(p) = \min_j \|z(y, p) - d_j\|_2^2. \tag{2}$$

The search covers the full memory, so the correspondence may change with location and view. The field a is the shared anomaly evidence, in the same patch-memory spirit as AnomalyDINO [4].

The image score upsamples a and averages the top fraction $\eta = 0.01$ of the resulting map. For the localization map, a 3×3 Gaussian-weighted average W ($h = 1$ in patch units, weights renormalized at borders) is applied on the native grid before upsampling. Because W preserves constants and cannot raise the neighborhood maximum, it redistributes peaks. The upper tail of the smoothed field is therefore not the upper tail of the raw field, and the two outputs are

$$s(y) = \mathcal{Q}_{0.01}\big(\mathcal{U}(a)\big), \qquad A_{\text{loc}}(y) = \mathcal{U}(Wa), \tag{3}$$

where $\mathcal{U}$ is bilinear upsampling and $\mathcal{Q}_{0.01}$ is the top-1% average. The image score does not depend on the localization bandwidth.

We use DINOv2-Giant with registers, averaging blocks $\{-18, \ldots, -12\}$, at 672×672 (48×48 patches). The memory stores 30 in-plane rotations of each support image. Weights stay frozen, readouts are fixed for clean and shifted queries, and there is no test-time adaptation.

3 Results

We follow the SubspaceAD protocol [5] with $k \in \{1, 2, 4\}$ normal references, five seeds, and shared reference sets. Baselines are SPADE [10], PatchCore [1], WinCLIP [2], IIPAD [3], AnomalyDINO, and SubspaceAD. Clean baseline numbers are taken from SubspaceAD; Shift is evaluated on ARAD-Compact. Table 1 reports 4-shot image and pixel AUROC as Clean | Shift. Ours is the five-seed mean (standard deviation at most 0.7 I-AUROC).

On clean images the detector matches the strongest baselines. Under the paired shift it keeps more of that accuracy. At 4-shot on MVTec-AD, SubspaceAD falls from 98.0 to 80.1 I-AUROC and our detector from 98.5 to 91.6. Pixel AUROC moves from 97.9 to 91.7 for SubspaceAD and from 98.2 to 96.0 for our map. On VisA the shifted I-AUROC is 65.9 against 78.4, and shifted P-AUROC is 92.1 against 97.0. IIPAD is the closest image-level baseline under shift and still trails on both datasets.

Table 1: 4-shot detection, Clean | Shift. Shift is the ARAD-Compact mixed panel. Ours is the five-seed mean.

	MVTec-AD		VisA	
Method	I-AUROC	P-AUROC	I-AUROC	P-AUROC
SPADE	84.8 — 55.7	92.7 — 56.9	81.7 — 52.2	96.6 — 60.1
PatchCore	88.8 — 75.2	94.3 — 87.0	85.3 — 68.8	96.8 — 89.3
WinCLIP	95.2 — 64.7	96.2 — 79.4	87.3 — 58.3	97.2 — 84.1
IIPAD	96.1 — 84.8	97.0 — 92.2	88.3 — 76.8	97.4 — 94.6
AnomalyDINO	97.7 — 80.1	97.2 — 90.6	92.6 — 68.6	98.2 — 94.2
SubspaceAD	98.0 — 80.1	97.9 — 91.7	94.7 — 65.9	98.4 — 92.1
Ours	**98.5 — 91.6**	**98.2 — 96.0**	**95.2 — 78.4**	**98.9 — 97.0**

Extra shots recover clean ranking more than shifted ranking. Our MVTec-AD I-AUROC goes from 97.4 | 90.8 at 1-shot to 98.1 | 91.5 at 2-shot and 98.5 | 91.6 at 4-shot. On VisA the clean score rises from 94.1 to 95.2, while the shift score stays near 79. SubspaceAD on VisA rises from 93.2 to 94.7 clean and falls from 68.1 to 65.9 under shift as k goes from 1 to 4. A larger clean memory does not decide whether the residual came from the object or from the imaging.

Table 2 holds the 4-shot references fixed and changes one choice at a time (three seeds, not pooled with Table 1). Sharing the raw field for both tasks leaves the image score unchanged and costs the map (97.4 | 95.0 against 98.2 | 96.0 P-AUROC on MVTec-AD). Sharing the pooled field recovers the map and does not help detection. On the same descriptors, a PCA subspace that keeps 99% of the variance is nearly costless when the image is clean and loses the shift: I-AUROC falls from 91.5 to 80.0 on MVTec-AD and from 78.3 to 65.6 on VisA. Smoothing that subspace map does not restore the image score. Squared ℓ_2 neighbors also beat cosine neighbors of the same memory (on VisA, +2.8 clean I-AUROC and +4.5 under shift). The shift result is carried by the match. Encoder scale is almost invisible on clean MVTec-AD, where every variant we tried already reaches 97–98 I-AUROC, and it is visible under ARAD-Compact: DINOv2-S and DINOv3-S/B stay near 82, DINOv3-L reaches 86.7, and DINOv2-Giant with registers retains 91.5 [11].

The operators are not interchangeable. A separate probe, not the Shift column, re-renders each test image from severity S1 to S5 for one operator at a time. Low light leaves I-AUROC essentially unchanged through S5. Motion blur is the steepest drop, from 96.4 to 89.5 on MVTec-AD and from 90.7 to 79.9 on VisA. Defocus stays above 94 and 88 on the two datasets. The mixed-panel gain is a gain on a budgeted mixture in which motion, noise, and overexposure remain the sensitive directions.

Table 2: Matched 4-shot controls, Clean | Shift. Readout placement, residual model, and frozen encoder are varied one at a time.

	Variant	I-AUROC	P-AUROC
Readout MVTec	Shared $\mathcal{U}(a)$	98.3 — 91.5	97.4 — 95.0
	Shared $\mathcal{U}(Wa)$	98.2 — 91.4	98.2 — 96.0
	Split (ours)	**98.3 — 91.5**	**98.2 — 96.0**
Residual MVTec	ℓ_2 1-NN	**98.3 — 91.5**	**98.2 — 96.0**
	PCA, 99% var.	97.5 — 80.0	98.2 — 92.3
Residual VisA	ℓ_2 1-NN	**95.1 — 78.3**	**98.9 — 97.1**
	PCA, 99% var.	94.6 — 65.6	98.7 — 91.8
Encoder MVTec	DINOv2-S	97.6 — 81.5	98.3 — 91.9
	DINOv3-L	97.9 — 86.7	98.2 — 95.2
	DINOv2-G-reg	**98.3 — 91.5**	**98.2 — 96.0**

4 Conclusion

A change in observation is not, by itself, a change in normality. ARAD-Compact scores the clean image and one controlled acquisition change as a pair. The training-free detector matches mean multi-layer DINOv2 patches to clean normal references, then keeps the image score on the raw upper tail and the map on a locally supported field. Under matched descriptors, nearest neighbors retain more accuracy under that shift than a high-variance PCA subspace, and spatial pooling does not repair the subspace miss. Additional support images raise clean ranking more than they close the gap. Motion blur and severe photometric change remain the harder operators. The open problem is to decide, for the same object, whether the residual was produced by the object or by the way it was imaged.

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